
ELEC6095 Smart Grid

History and Development of Modern Grid

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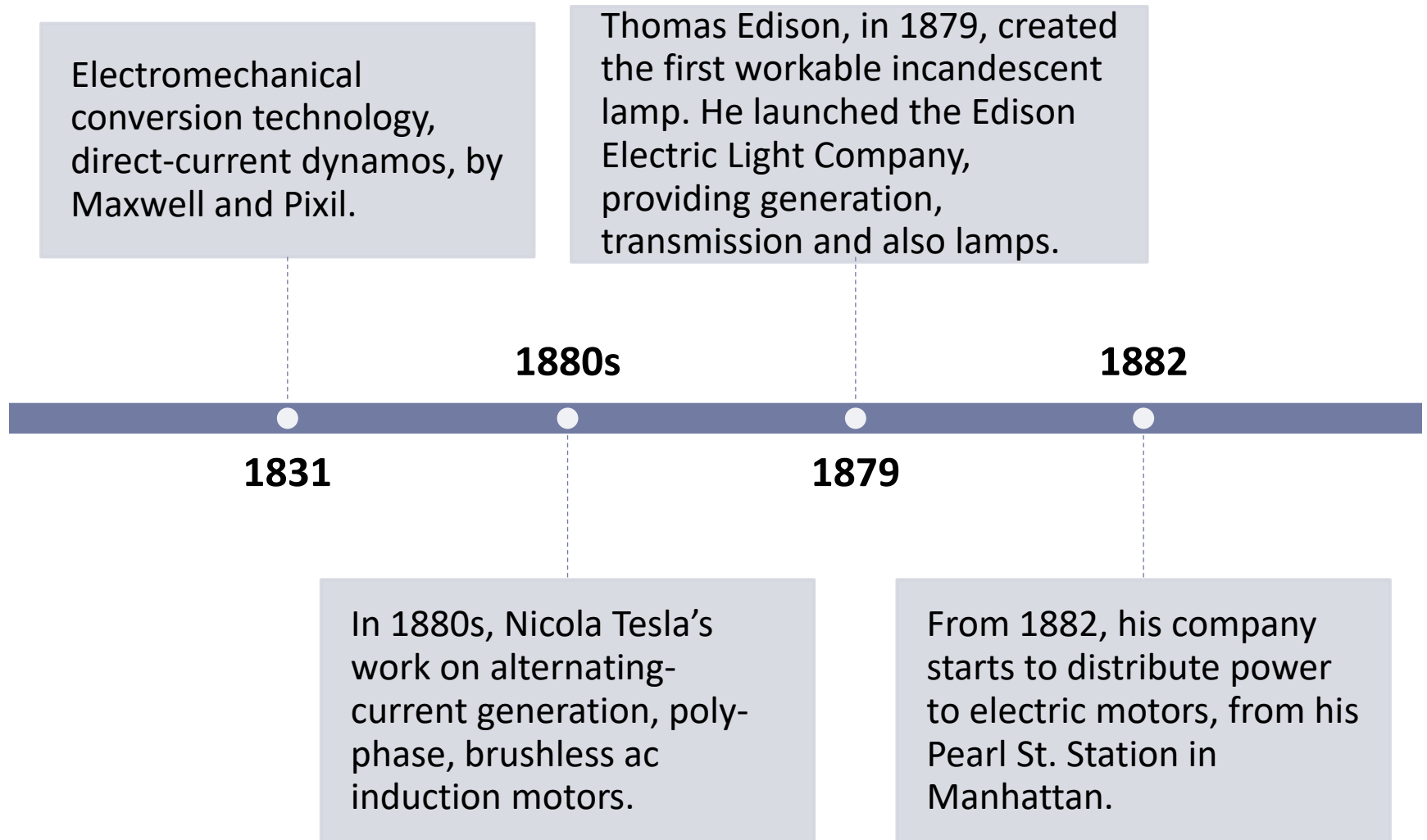
Power Industry Revolutions



In 1880

- ▶ Just more that 100 years ago, there were no light bulbs, refrigerators, air conditioners...,
- ▶ Also, there was no pollution.
- ▶ In late 19th century, scientific descriptions of electricity and magnetism, by Hans Christian Oersted, Andre Marie Ampere, and James Clerk Maxwell.
- ▶ On the centenary of Maxwell's birthday, his work was described by Einstein as the "most profound and the most fruitful that physics has experienced since the time of Newton".

Invention of electrical power generation



Edison's generator

- ▶ Edison's system was based on **direct current (DC)**, advantages:
 - ▶ it provides flicker-free light
 - ▶ enables easier speed control of DC motors.
- ▶ The disadvantages of DC of those days were:
 - ▶ It was difficult to change voltage level in DC.
 - ▶ After the invention of transformer in 1883, it is easy to do it in AC.
 - ▶ Edison's customers had to be located within just one or two km of a generating station.
 - ▶ High voltage level can help reduce losses.

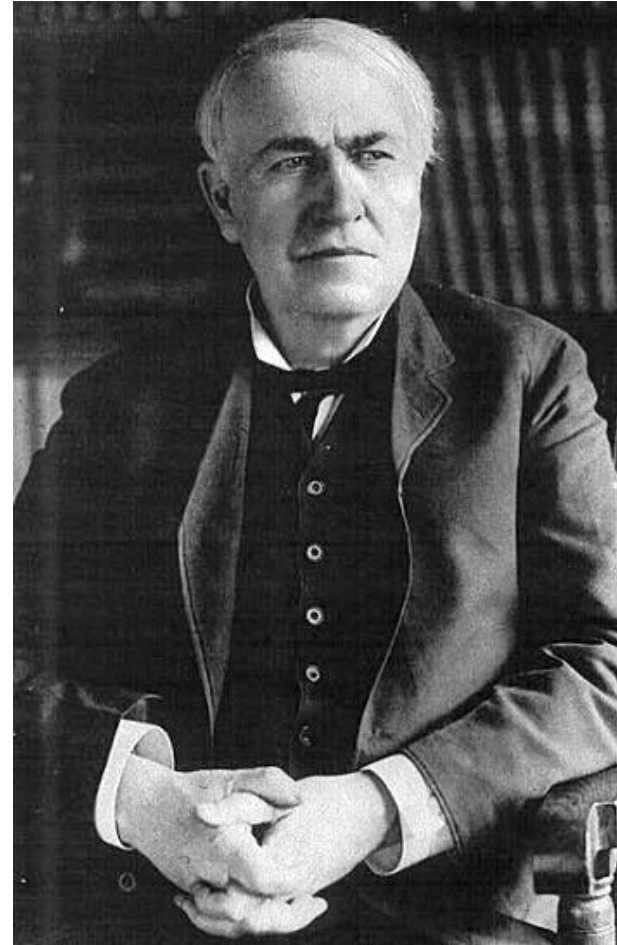
Westinghouse and Edison

- ▶ **George Westinghouse** recognized the advantages of ac for long distance transmission, he utilized ac technology developed by Tesla, and launched the Westinghouse Electric Company in 1886.
- ▶ **Edison** tried to prove that the high voltage of AC is a safety hazard → result is the development of electric chair.
- ▶ **Westinghouse** competed with Edison. Then, AC won.
- ▶ The Current War

AC won in 1880s

In the battle of DC and AC, the advantages of high-voltage transmission won.

Through buyouts and mergers, Edison incorporated General Electric Company in 1892, focus on manufacturing electrical equipment and end-use devices.



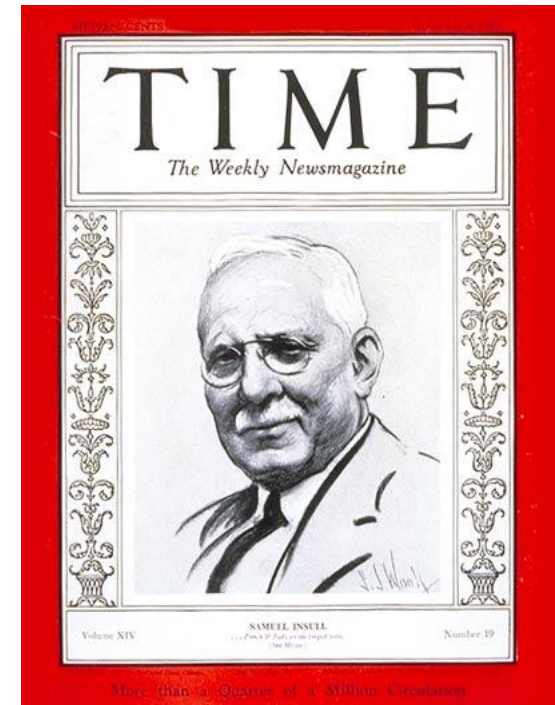
Thomas Edison

Samuel Insull's contributions to power industry

Samuel Insull devised the industry structure and business model for power industry.

A regulated, vertically integrated system.

The business model enabled the power industry for a continuously growth over a century.



Samuel Insull's contributions to power industry

- ▶ Electricity was a luxury.
- ▶ Samuel Insull's visions
 - ▶ It was cheaper to serve customers when their power use was aggregated, *mass consumption*.
 - ▶ Economies of scale in production
 - ▶ Regulation can protect an industry (power industry) with declining costs, and high capital needs.
- ▶ Samuel Insull's contributions:
 - ▶ Costs go down and supply goes up → sell more and charge less
 - ▶ Economies of scale
 - ▶ Recognized the inefficiencies associated with multiple power companies competing for the same customer. Established the concept of regulated monopolies.

Power Industry before 1990s

- ▶ Insull's visions
 - ▶ Large supply
 - ▶ Massed demand
 - ▶ Increased consumption
 - ▶ Regulated rates
- ▶ Insull's visions have dominated power industry development. Without it, electrification might not have happened.

Power Industry before 1990s

- ▶ Almost all power systems in the world were well developed
 - ▶ to serve electricity in a cheaper way → coal fired power plants
 - ▶ to serve aggregated customers (mass demand) → large scale power plant, centralized generation
- ▶ The regulated large-scale centralized power generation based traditional vertically integrated power systems worked well since 1880s until recent decade.
- ▶ It is facing new challenges, which are far larger than any in its history.

The Second Electric Revolution

The challenges are motivated by two worldwide policy imperatives:

- The need to adopt policies reducing the impacts of global climate changes.
- The need for energy security (energy crisis, oil import).

The industry need to:

- Change fossil-fuel based generation to renewable and clean energies.
- Improve energy efficiency.
- Adopt electric vehicles to reduce the rely on oil.

Challenges to power industry



The industry have to make a massive changes in its fuel sources and generating plants at a wholly unprecedented pace.



The challenge is not simply swapping out old plants for new ones.



For example, a 1 million MW fossil fuel based system will require a trillion US\$ retooling in the next several decades.

Smart Grid

- ▶ As the industry shifts its supply sources, builds transmission, and increases energy efficiency, **the technologies at the core of its operations** will shift dramatically
- ▶ The industry will adopt the **Smart Grid**.
- ▶ The architecture of the system will shift from **the one based on large sources and central control** to **the one with many more smaller sources and decentralized intelligence**.
- ▶ This is the first major architectural change since AC was adopted in 1893.

New System – Smart Grid

- ▶ The new system have to be designed with three objectives:
 1. Creating a decentralized control paradigm;
 2. Retooling the system for low-carbon supplies; and
 3. Finding a business model that promotes much more efficiency.

Creating a decentralized control paradigm

- ▶ **Distributed Intelligence:** embedding computing capability throughout the network, from substations to smart meters and home energy management systems.
- ▶ **Edge Computing Architectures:** processing data locally where it's generated to enable faster decision-making without constant central oversight.
- ▶ **Multi-Agent Systems:** autonomous software agents that negotiate and coordinate with each other to optimize local resources while maintaining system stability.
- ▶ **Self-Healing Capabilities:** automated fault detection, isolation, and service restoration without central intervention.
- ▶ **Interoperability Standards:** common protocols enabling diverse devices from different manufacturers to communicate seamlessly.

Retooling the system for low-carbon supplies

- ▶ **Bidirectional Power Electronics:** advanced inverters and controllers that manage variable generation and complex power flows.
- ▶ **Dynamic Thermal Ratings:** systems that monitor actual transmission line capacity rather than using conservative static ratings, increasing renewable hosting capacity.
- ▶ **Synthetic Inertia:** programming inverter-based resources to mimic the stabilizing effects traditionally provided by fossil fuel generators' rotating mass.
- ▶ **Weather Forecasting/Nowcasting Integration:** Direct coupling of meteorological data with grid operations to anticipate renewable energy fluctuations.
- ▶ **Sector Coupling:** integrating electricity with heating, transportation, and industrial energy use to maximize decarbonization potential.

Business model that promotes efficiency

- ▶ **Transactive Energy Frameworks:** market structures that enable automated, granular trading of energy services between consumers, prosumers, and utilities.
- ▶ **Value-Stacking Mechanisms:** allowing a single asset (like battery storage) to provide and be compensated for multiple grid services simultaneously.
- ▶ **Outcome-Based Regulation:** shifting utility compensation from infrastructure investment to performance metrics like efficiency, reliability, and carbon reduction.
- ▶ **Aggregator Business Models:** enabling third parties to bundle distributed resources and participate in wholesale markets.
- ▶ **Data Monetization:** creating value from the massive data generated by smart meters and sensors while respecting privacy and security concerns.



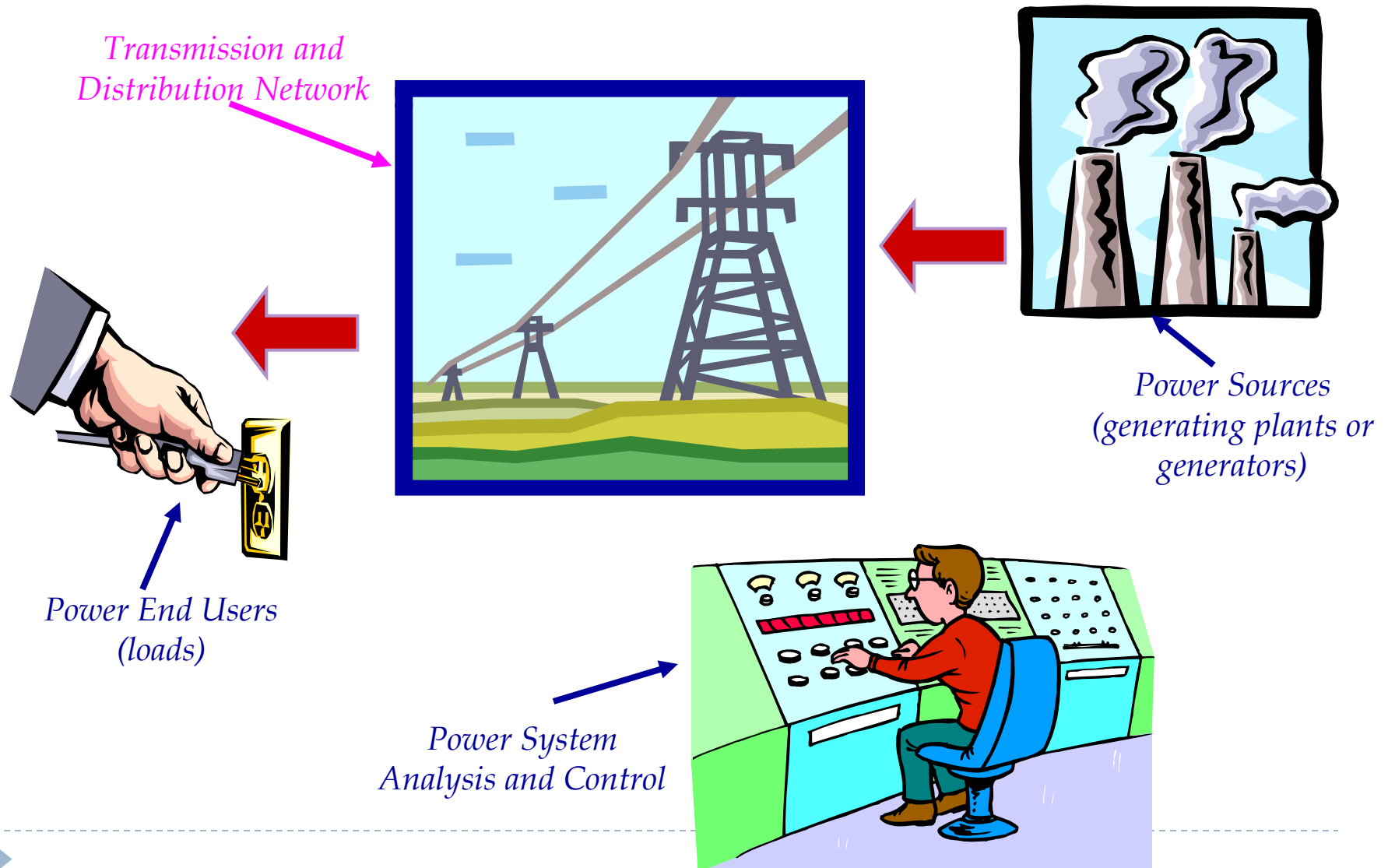
Power System Basics



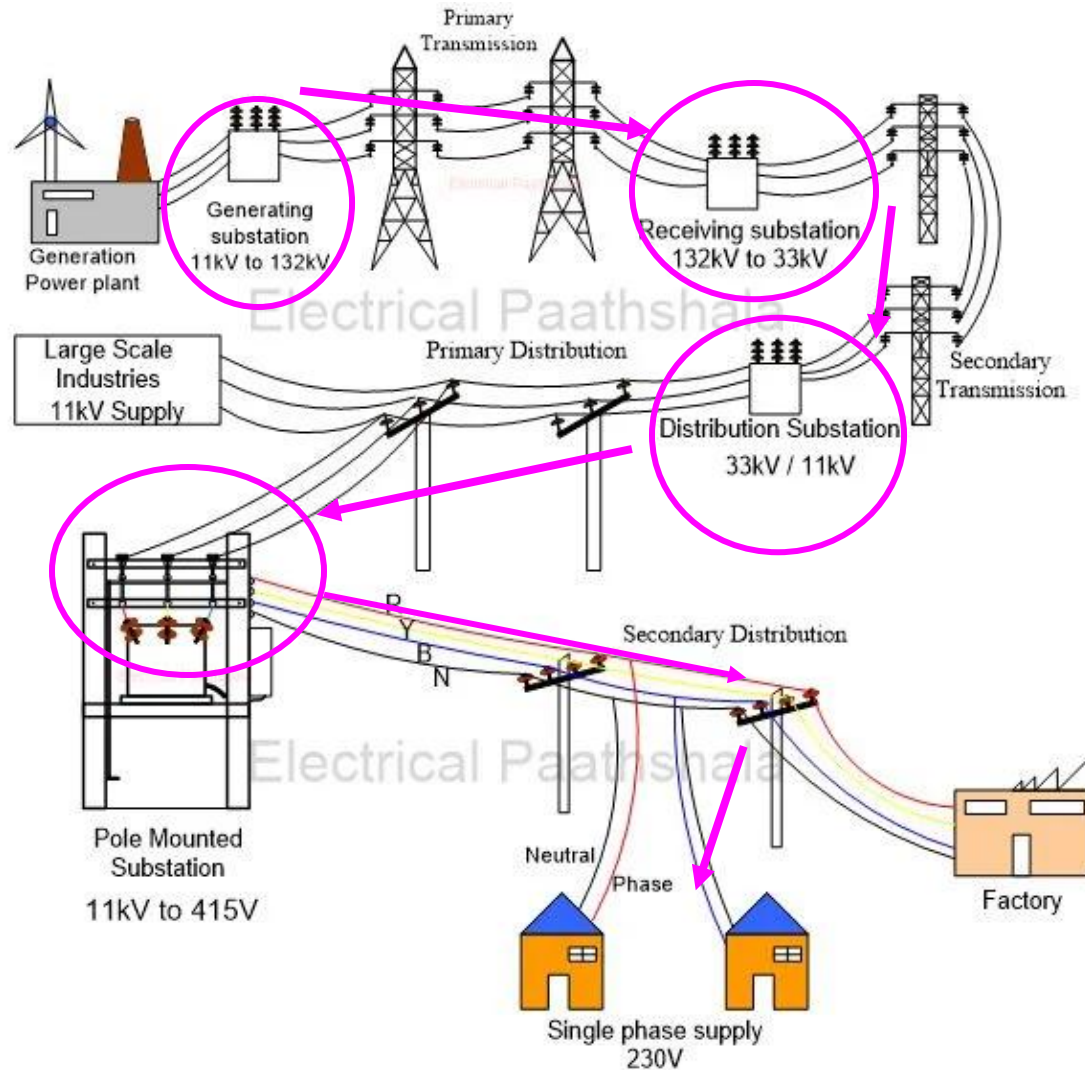
Power Systems

- ▶ Use synchronous machines for generation of electricity.
 - Primary sources of energy (fossil fuel, nuclear, and hydraulic) → Mechanical energy → electrical energy
- ▶ Transmit power over long distances to customers spread over a wide area.
- ▶ Distribute electricity to small industries and local residents

Power Systems



How do we get electricity at home?

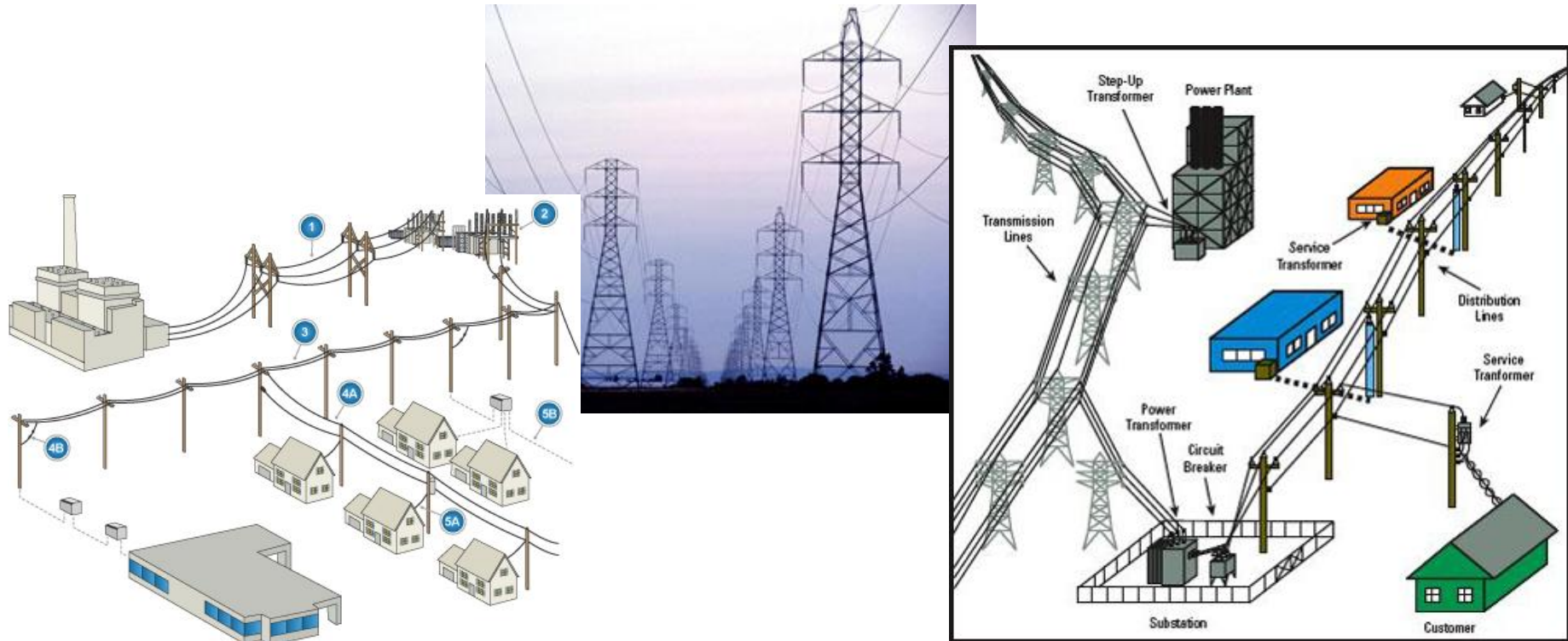


Generators

- ▶ Most commonly, the generating plants convert primary energy into electrical energy.
 - Fossil fuels (coal, petroleum, gas), or
 - nuclear fuels, or
 - falling water.
- ▶ Renewable Energy (RE) sources are widely used in recent years.
 - Wind power, solar energy, geothermal, biomass energy, etc.

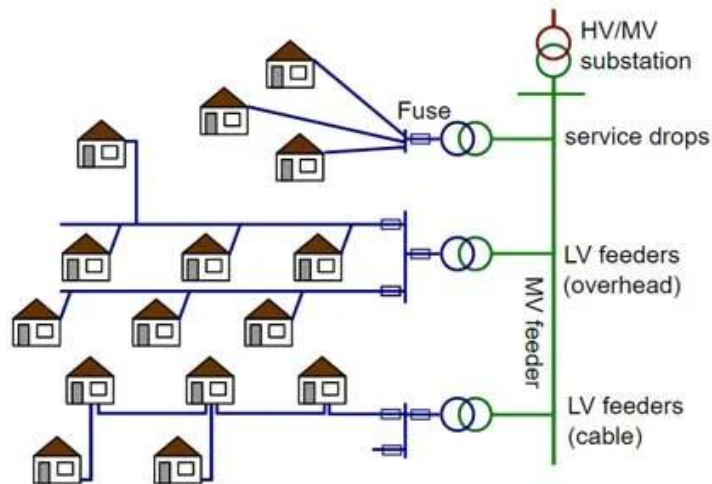
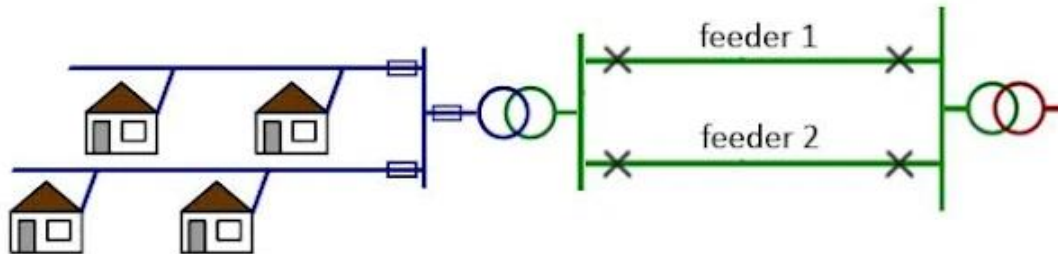
Electricity Transmission & Distribution

- ▶ Generation sources are usually interconnected by a transmission system or a network that distributes the power to various load centers.

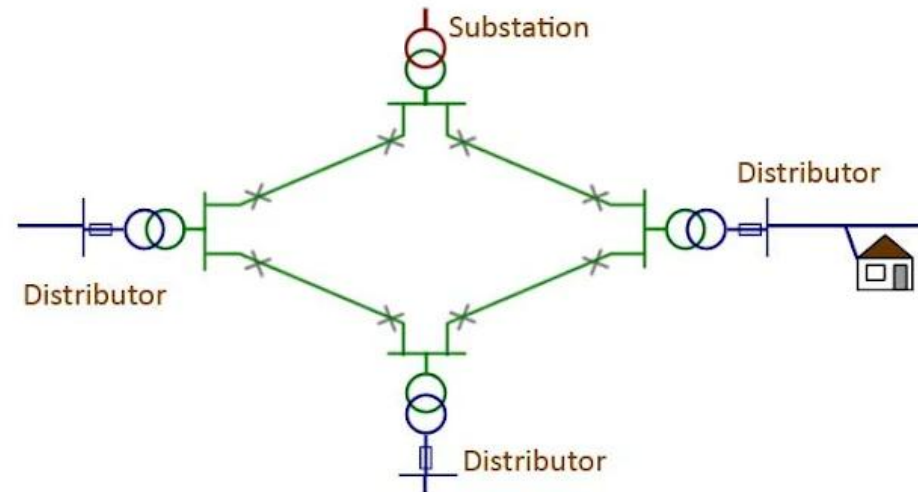


Distribution systems

Parallel feeders



Radial distribution



Ring distribution

Distribution systems worldwide

- ▶ Distribution systems around the world have evolved into different forms: North American and European.
- ▶ For both forms, hardware is almost the same: conductors, cables, insulators, arresters, regulators, and transformers are very similar. Distribution voltages and power carrying capabilities are also similar.
- ▶ Secondary voltages have driven many of the differences in distribution systems.
- ▶ North America has standardized on a 120/240 V secondary system; on these, voltage drop constrains how far utilities can run secondaries, typically not more than 100m.
- ▶ In European designs, higher secondary voltages allow secondaries to stretch to almost 1.5km. European secondaries are largely three phase and most European countries have a standard secondary voltage of 220, 230, or 240 V, twice the North American standard.
- ▶ With twice the voltage plus three-phase secondaries, a European distribution design can reach at least 8 times the length of an American design for a given load and voltage drop.



Deregulation



-- the transition to Smart Grid

Deregulation of Power Industry



In 1990, the future of the power industry could be summarized by “*deregulation and market*”



Follow the deregulation of airlines, telephone companies, natural gas suppliers...



20 years later, the popular topics of power industry are climate change, energy efficiency, and smart grid.

Deregulation of Power Industry

- ▶ Deregulation and market mechanisms are different in different countries
- ▶ Bilateral contracts, zonal pricing based
- ▶ Nodal pricing based
- ▶ Deregulate the market to customers, retail market
- ▶ The target was to introduce competitions in generation side, and hence reduce electricity retail prices
- ▶ First energy crisis in 1970s
- ▶ Open customer side is necessary, → Demand Response (DR), Load Management (LM), microgrid, etc.

Electricity billing

- ▶ Since a long time ago, electricity customers are charged by energy measured in kWh.
- ▶ The power company reads customer meter every month.
- ▶ The kWh used is measured according to cumulative monthly use.
- ▶ This is the only way can be done since the power industry was originally developed, no WiFi, no sensors, no IT...

- ▶ Nowadays, digital meters, WiFi, sensors, IT,
- ▶ And Electricity Pricing
- ▶ They open up a panorama of new possibilities....

New possibilities



Setting time-based prices and billing them becomes much easier.



Smart meters can work with smart appliances to automatically response to price changes.



Smart meters make it easier to integrate small-scale generators and storages from customer sides.

Power system operations with new possibilities

- ▶ Power system operation

- ▶ Real-time balancing is the key
- ▶ In the past, real-time balancing is implemented from the generation side; generation scheduling, balancing, frequency regulation...

- ▶ Smart meters and time-based pricing create a new operating paradigm

- ▶ In addition to real time balancing from generators, the users of power systems self-balance the system by adjusting their own use.

Smart Grid from transmission viewpoints

- ▶ In the transmission systems, the impact of smart technologies is quite different.
- ▶ Smart grid technologies are better in monitoring the status of all power lines.
- ▶ Forecast reliability problems.
- ▶ Diagnose the problem when a power line trips.
- ▶ Make quick decisions when fault events happen.

Smart Grid from transmission view points

- ▶ Control system / control center in a smart grid could be compared to a control system of an airplane, which is
 - ▶ Packed with sophisticated gear (as to be nearly impossible to operate by human skill alone)
 - ▶ Relied on sensors and automatic control agents that obtain information quickly and act accordingly
 - ▶ Pilot use sensing parameters, etc., for stabilizing the aircraft.
- ▶ Power Grid control has similar concept
 - ▶ *Situational Awareness* of the grid
 - ▶ It allows fast-acting changes in power production and power routing.

Energy storage in Smart Grid

- ▶ With communication technologies, and bi-directional communication between utilities and customers, it is easier to accommodate:
 - ▶ Energy storage devices
 - ▶ Electric Vehicles
- ▶ They could be switched on / off quickly
- ▶ Provide real-time power balancing
- ▶ etc.



The New Paradigm

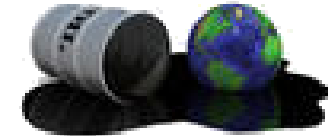


overview of Smart Grid technologies

Challenges to Electrical Power Grid



Increasing demand
for electricity



Energy crisis
Environmental issue

Higher requirements on
reliability, security and efficiency

Aging infrastructures



Energy and Environmental Issues

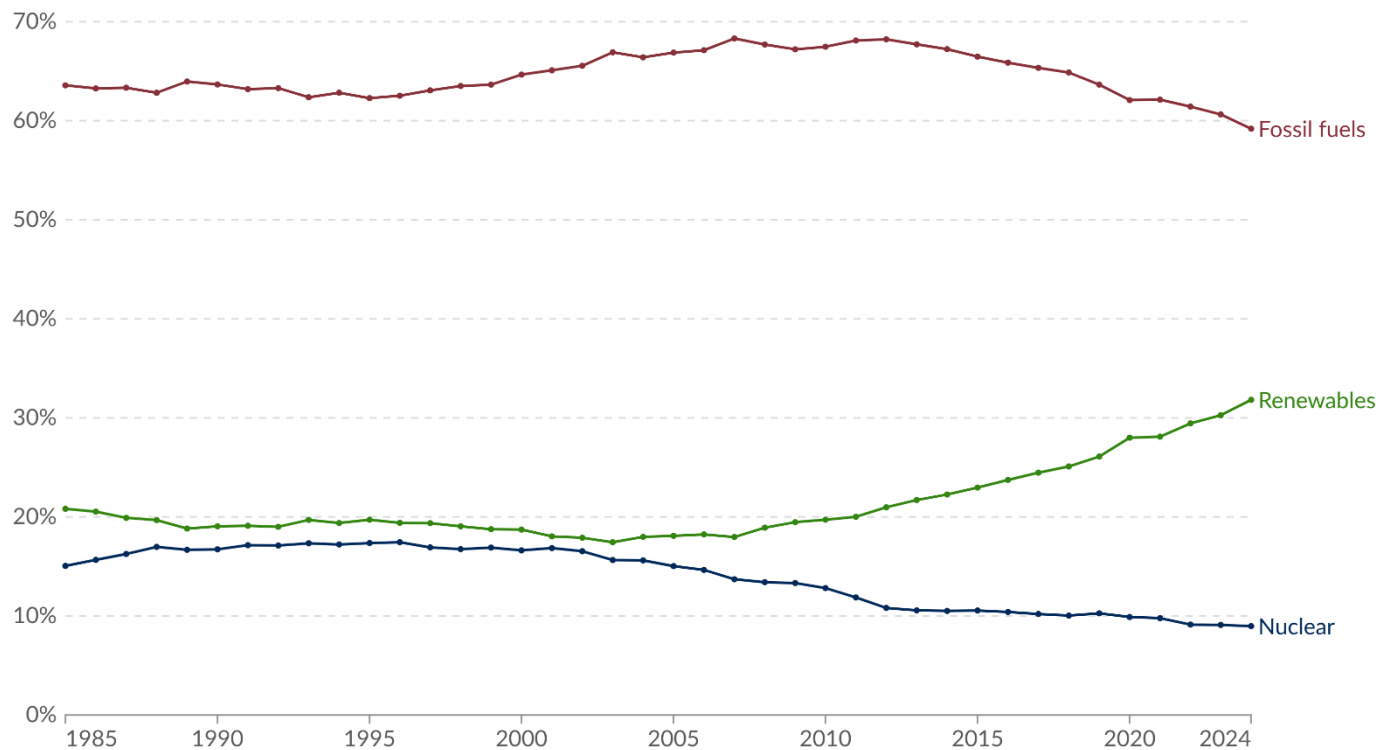
- ▶ Fossil fuels have been the primary energy.
- ▶ More than 60% of electricity **WAS** from fossil fuel based power plants.
- ▶ Issues of fossil fuel generation
 - ▶ Carbon emission, SOx, pollution, global warming, and energy depletion, etc...
- ▶ Solutions:
 - ▶ Install clean energy and renewable energy based generation to accommodate fast load growth.
 - ▶ Demand management using IT.

Share of Renewable Energy (the World data)

Share of electricity generation from fossil fuels, renewables and nuclear, World

Our World
in Data

Measured as a percentage of total electricity produced in the country or region. Fossil fuels include coal, oil, and gas. Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal.



Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025)

OurWorldinData.org/energy | CC BY

Source:

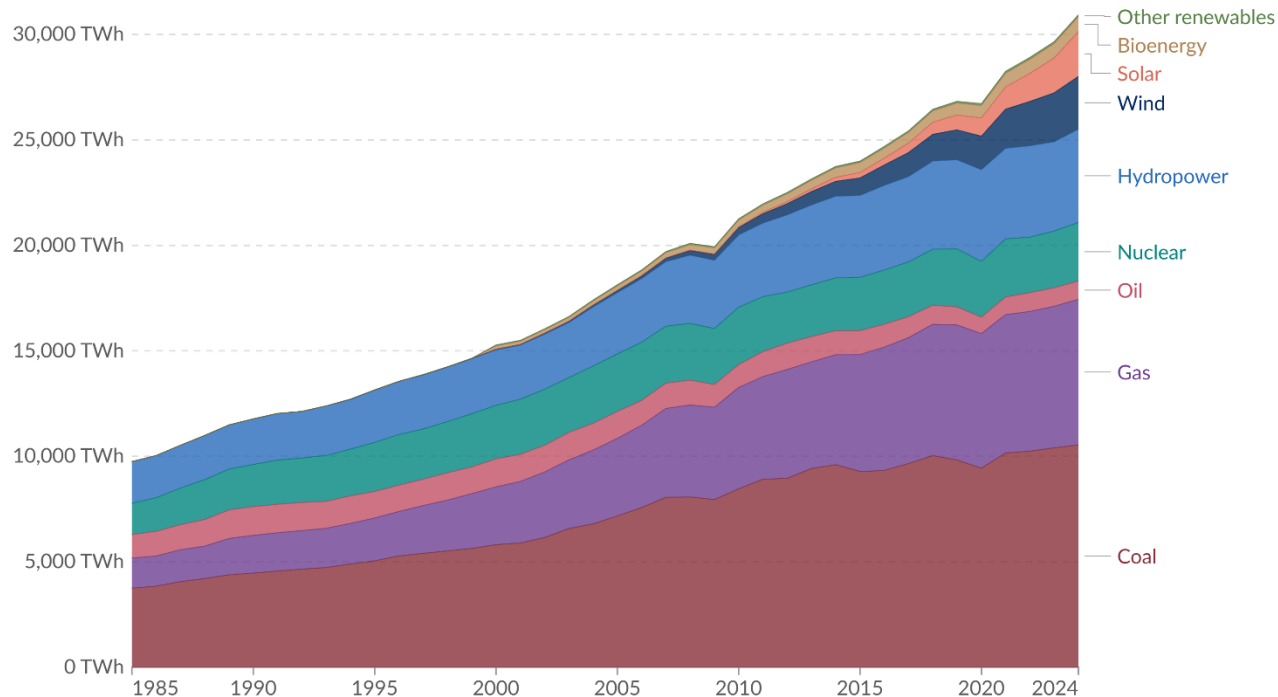
<https://ourworldindata.org/electricity-mix>

World gross electricity production by source

Electricity production by source, World

Measured in terawatt-hours¹.

Our World
in Data



Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025)

OurWorldinData.org/energy | CC BY

Note: "Other renewables" include geothermal, wave, and tidal.

Source:
<https://ourworldindata.org/electricity-mix>

1. Watt-hour A watt-hour is the energy one watt of power delivers for one hour. Since one watt equals one joule per second, a watt-hour equals 3600 joules of energy.

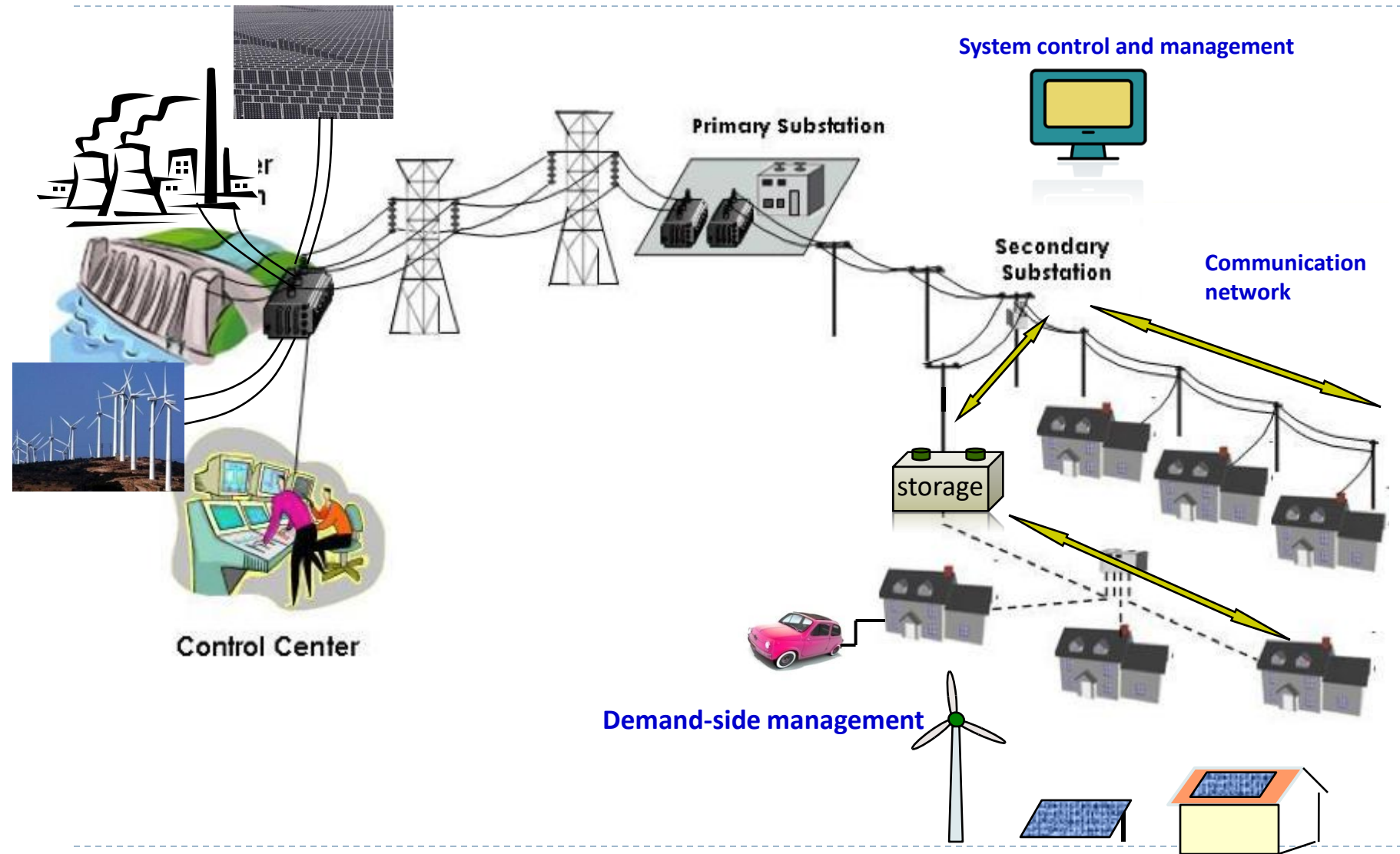
Metric prefixes are used for multiples of the unit, usually:

- kilowatt-hours (kWh), or a thousand watt-hours;
- Megawatt-hours (MWh), or a million watt-hours;
- Gigawatt-hours (GWh), or a billion watt-hours;
- Terawatt-hours (TWh), or a trillion watt-hours.

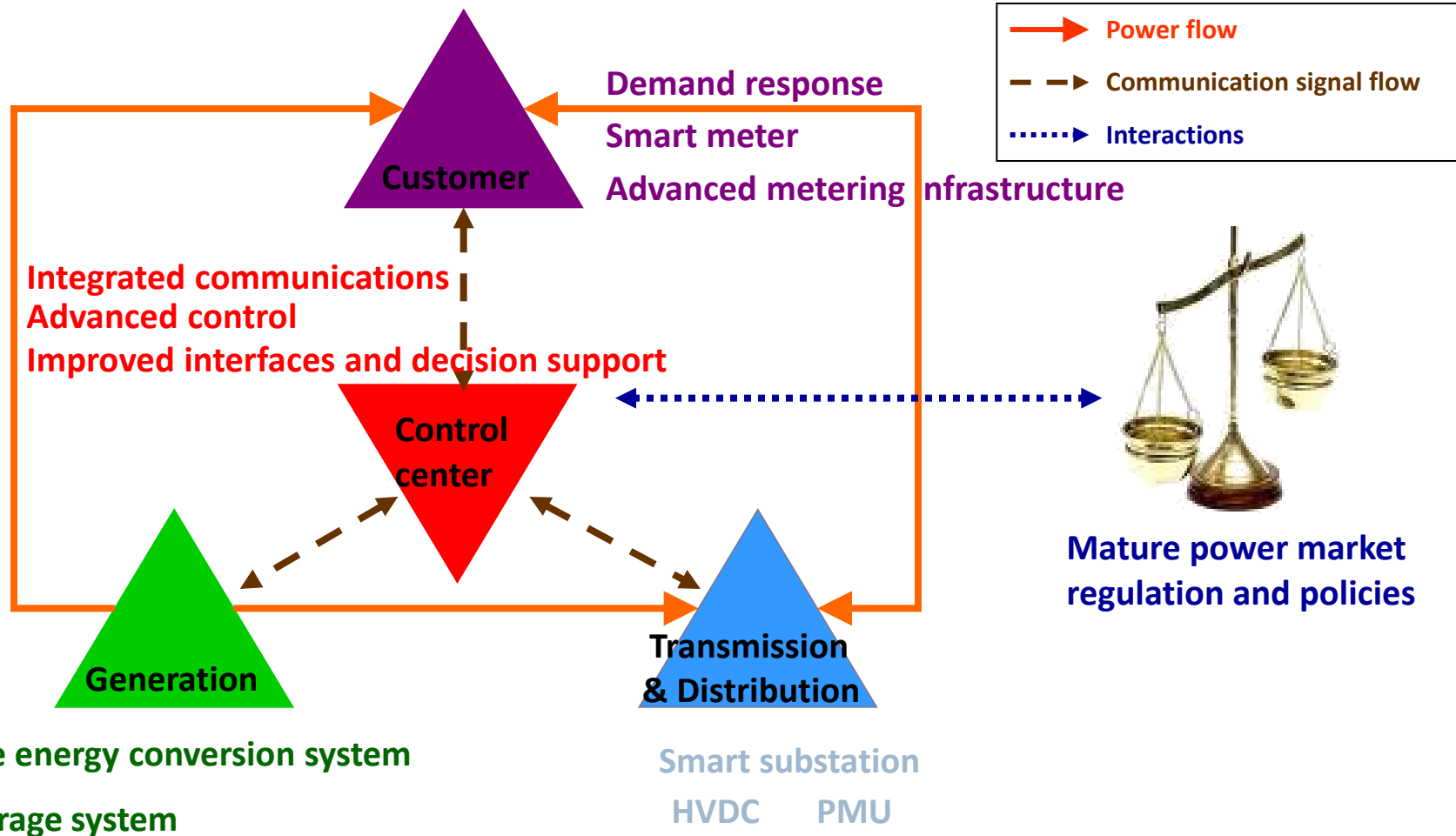
Renewable Energy grid integration

- ▶ High penetration of renewable energy generation in the power grid will cause some issues:
 - ▶ Intermittent renewable energy, uncontrollable.
 - ▶ Uncertainties in real-time power balance due to intermittent RE.
 - ▶ Power system security operation problem.
 - ▶ Determine RE penetration level.
 - ▶ Smart demand side management.
 - ▶ Unpredictable changes in energy demand.

Smart Grid with Renewable Energy



Smart Power Grid - Functions



Smart Customers

- ▶ Smart Customers are electricity consumers who actively participate in grid operations through technology-enabled engagement and decision-making:
 - ▶ Using smart technologies and devices (smart meters, smart appliances, home energy management systems).
 - ▶ Responding to price signals and demand response programmes.
 - ▶ Potentially generating their own electricity (becoming "prosumers").
 - ▶ Adjusting their consumption patterns based on grid conditions.
 - ▶ Participating in energy markets and programmes.



Smart Meter

Functions:

- ▶ Measure and record electricity consumption.
- ▶ Enable consumers to participate in demand response programs.
- ▶ Provide data that address electricity service issues.



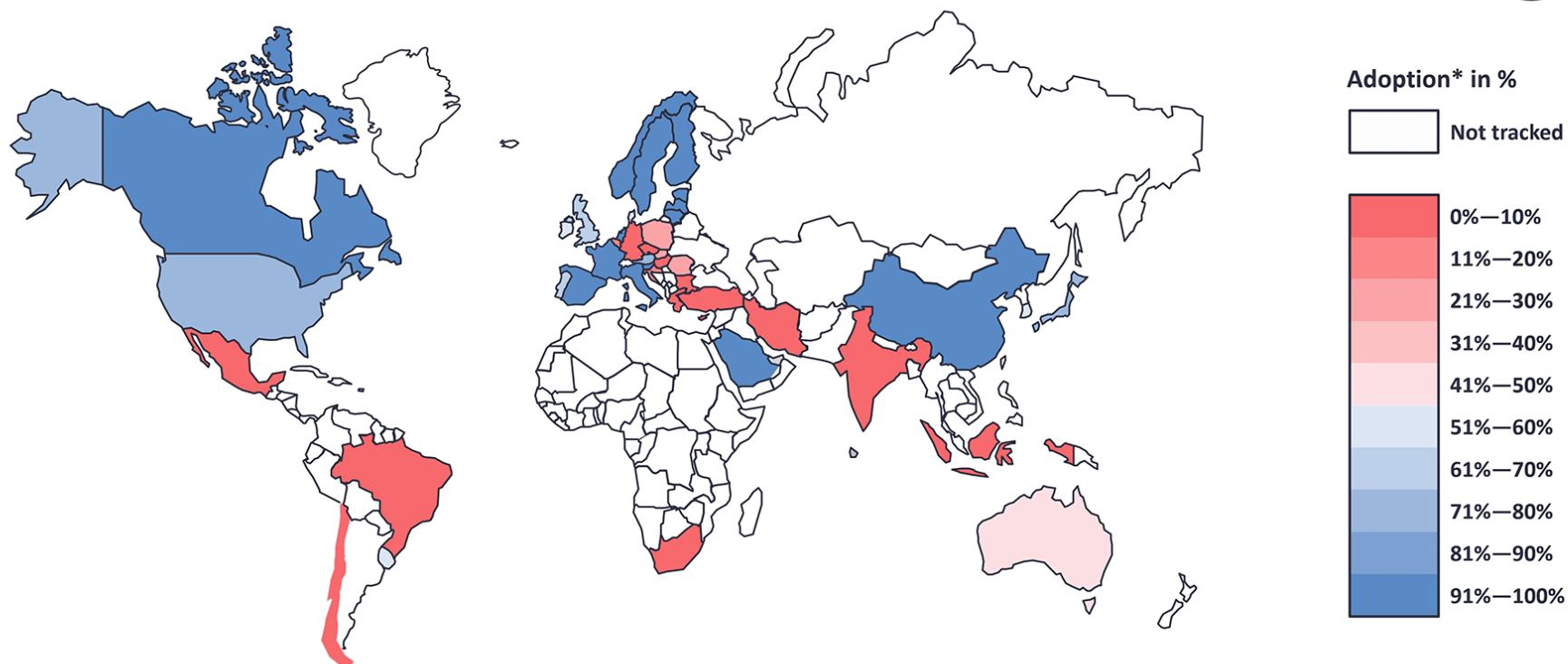
Basic component of
Advanced Metering
Infrastructure (AMI)

Core technology - communication

- ZigBee
- Power-line Communication (PLC)
- General Packet Radio Services (GPRS)
- Hybrid Fiber-Coaxial (HFC) bidirectional communication



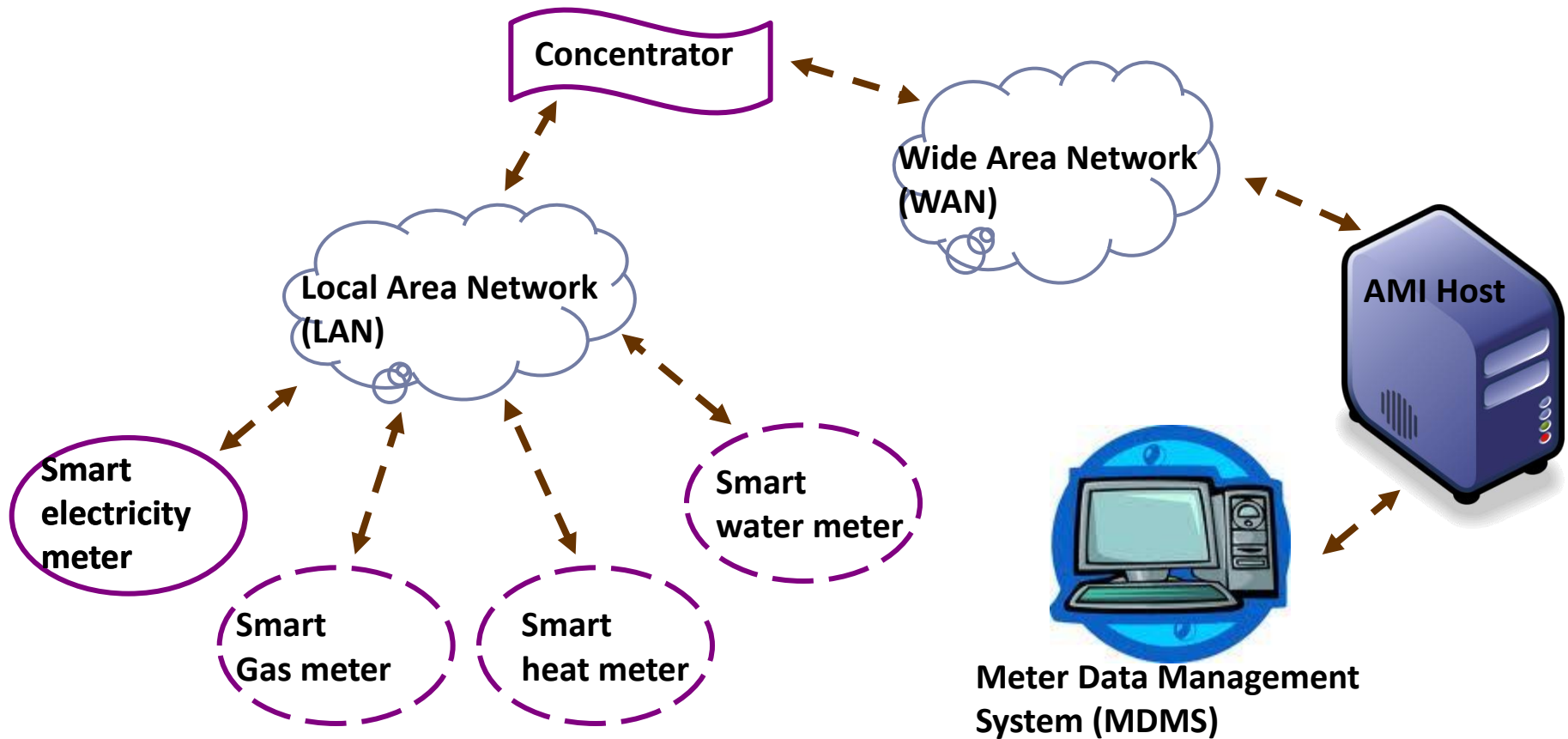
Global smart electricity meter adoption 2024



* =Smart meter adoption (as of Q4 2023) = The share of smart (AMI) electricity meters installed as a share of the total number of metering points that includes residential, commercial, and industrial segments.
 Source: IoT Analytics Research 2024 – Global Smart Meter Market Tracker 2020-2030. We welcome the republishing of images but ask for source citation with a link to the original post or company website.

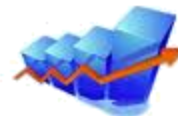
Advanced Metering Infrastructure (AMI)

AMI ≠ AMR (Automatic metering reading)



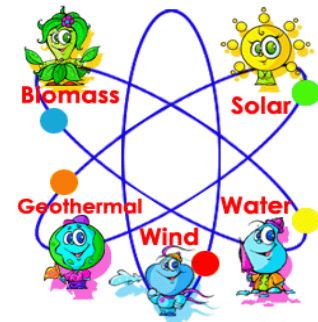
Demand Response

- ▶ Aim to manage customer consumption of electricity in supply conditions.
 - ▶ Emergency demand response
 - ▶ Economic demand response
 - ▶ Environmental demand response
- ▶ Prerequisites
 - ▶ Real-time pricing mechanism
 - ▶ Energy demand management
 - ▶ Incentive policies



Smart Generation

- ▶ Smart Generation refers to electricity production systems that are intelligently integrated into the grid infrastructure with bidirectional communication and control capabilities:
 - ▶ Generation sources equipped with advanced monitoring and control technologies that communicate with broader grid network.
 - ▶ Ability to adjust output dynamically in response to real-time grid conditions, demand signals, and electricity prices.
 - ▶ Integration of renewable energy sources (solar, wind, etc.) with predictive forecasting capabilities to manage intermittency.
 - ▶ Distributed energy resources that can be aggregated and coordinated across the network.
 - ▶ Self-optimization capabilities to maximize efficiency, profitability, and environmental benefits.
 - ▶ Fault prediction and resilience features that help prevent outages or minimize their impact.



Renewable Energy

► Pros:

- Energy cost saving
- No / less pollution

► Cons:

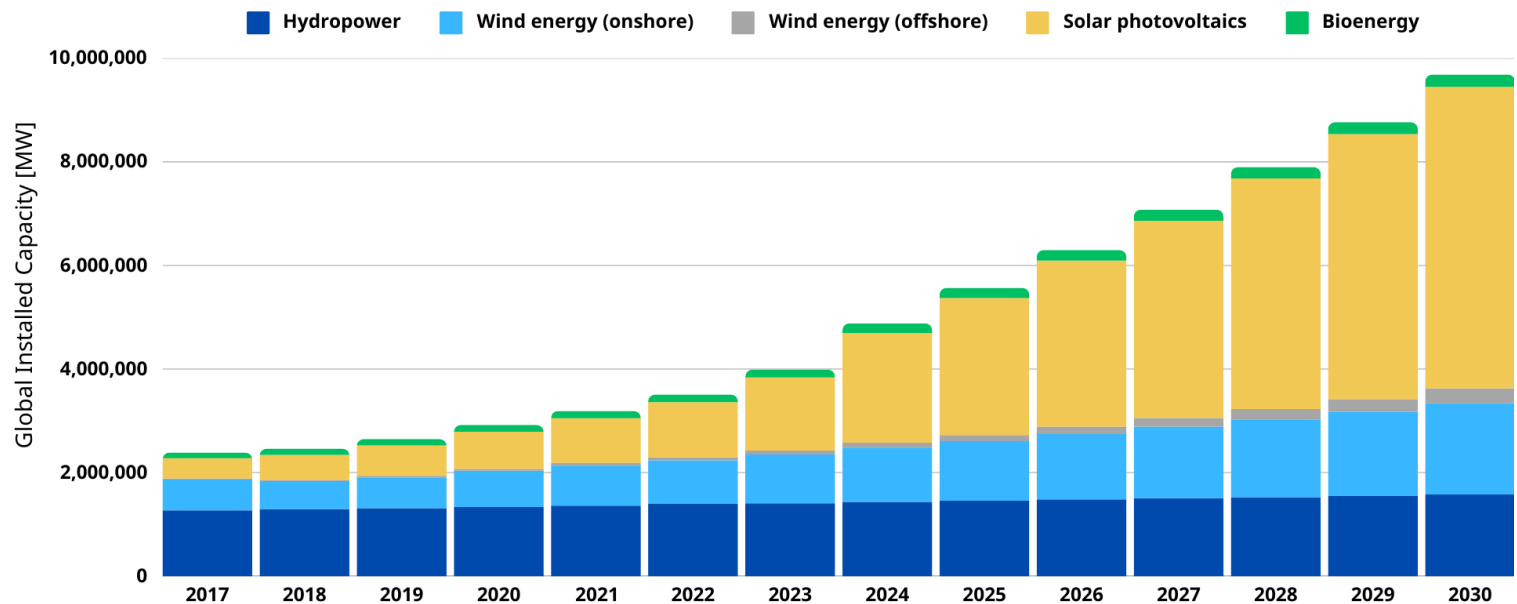
- Intermittency
- Uncontrollability
- Location requirements



Total renewable capacity worldwide (GW)

Renewable Energy Capacity Growth

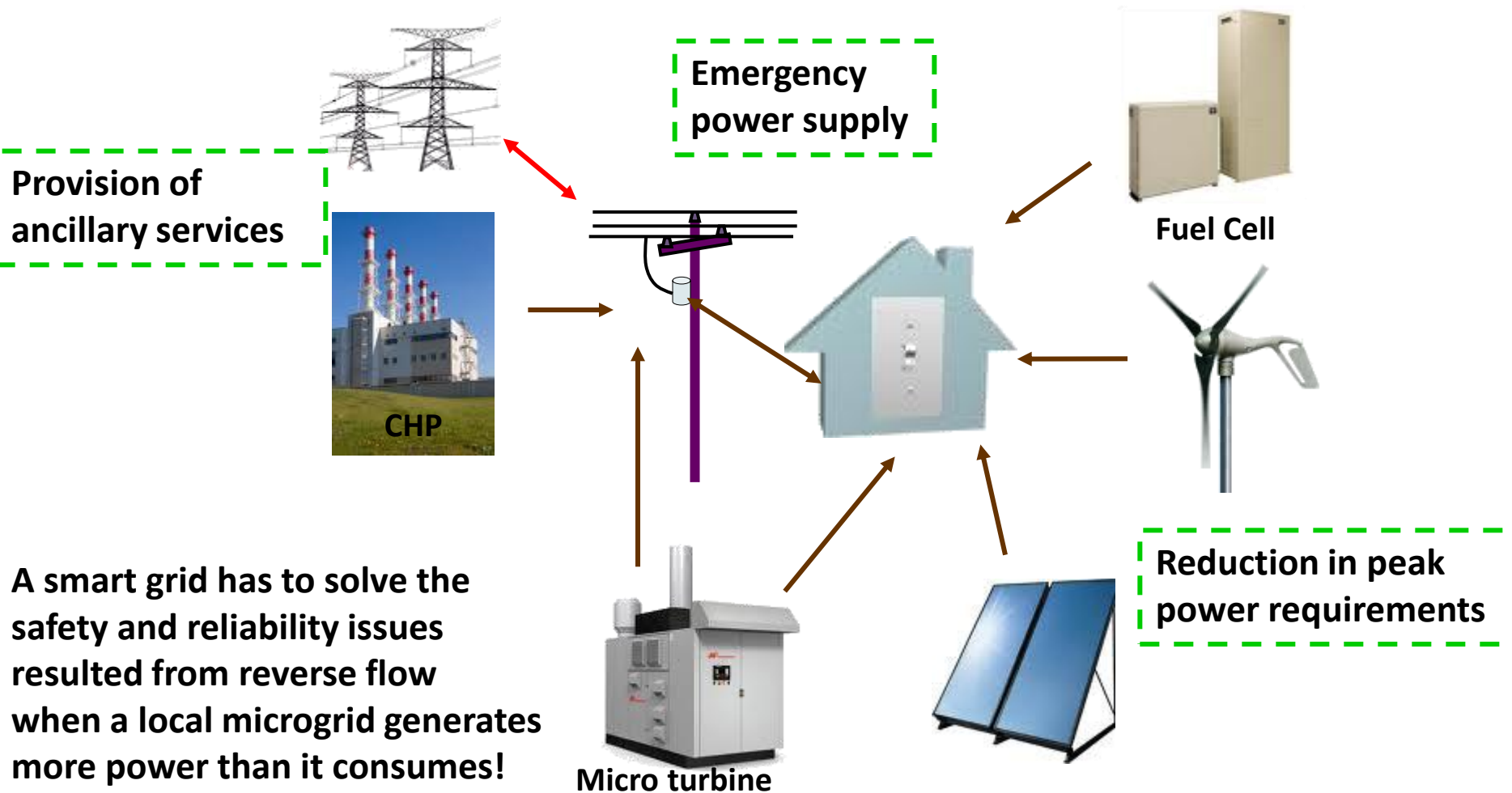
2017-2023 Installed Capacity and 2024-2030 IEA Forecast



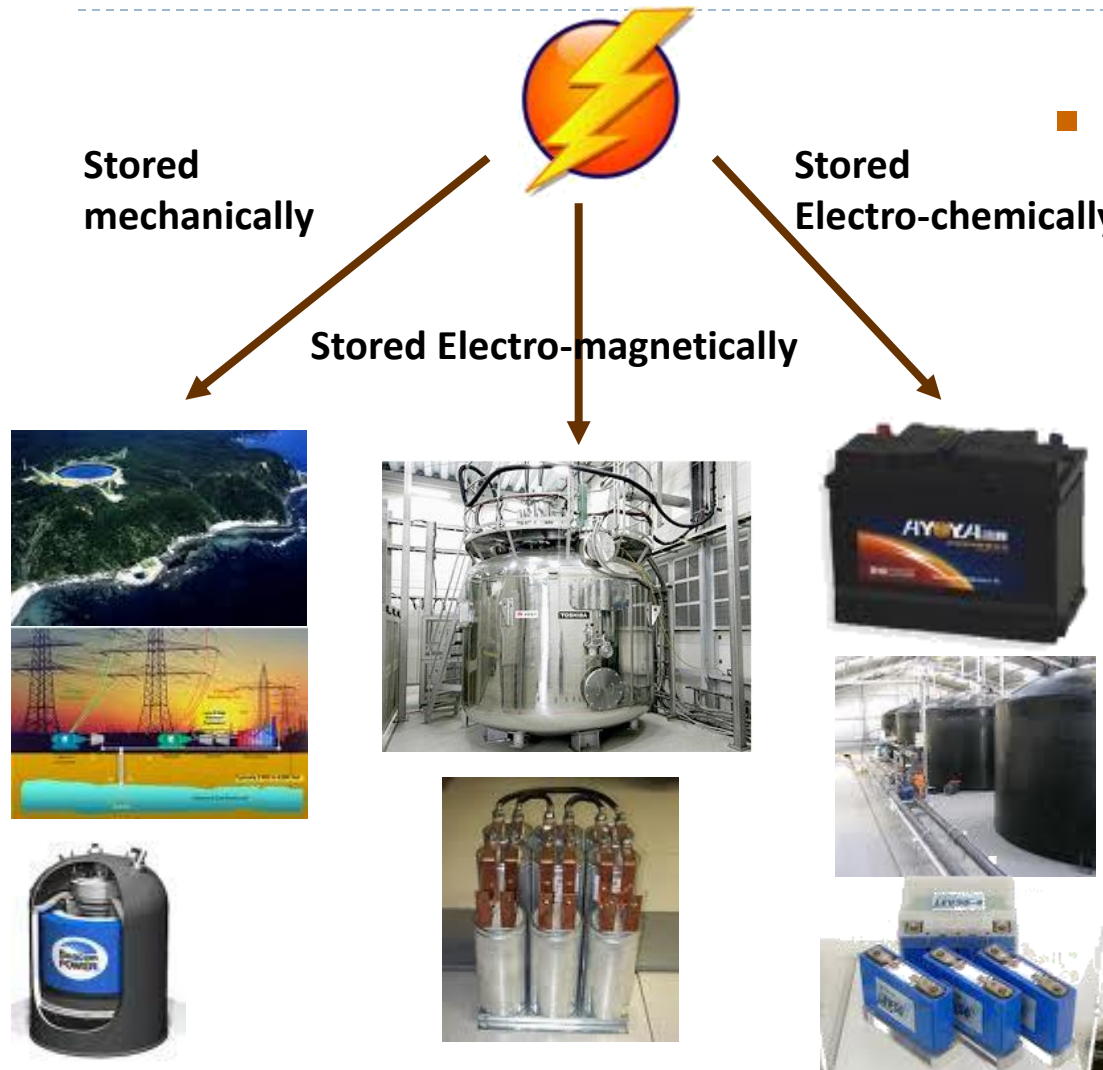
Source:

<https://rejobs.org/en/renewable-energy-blog/renewable-energy-forecast-for-2030>

Distributed Generation



Energy Storage System



■ Categories

» Store Energy Mechanically

- Pump storage
- Compressed air energy storage (CAES)
- Flywheels

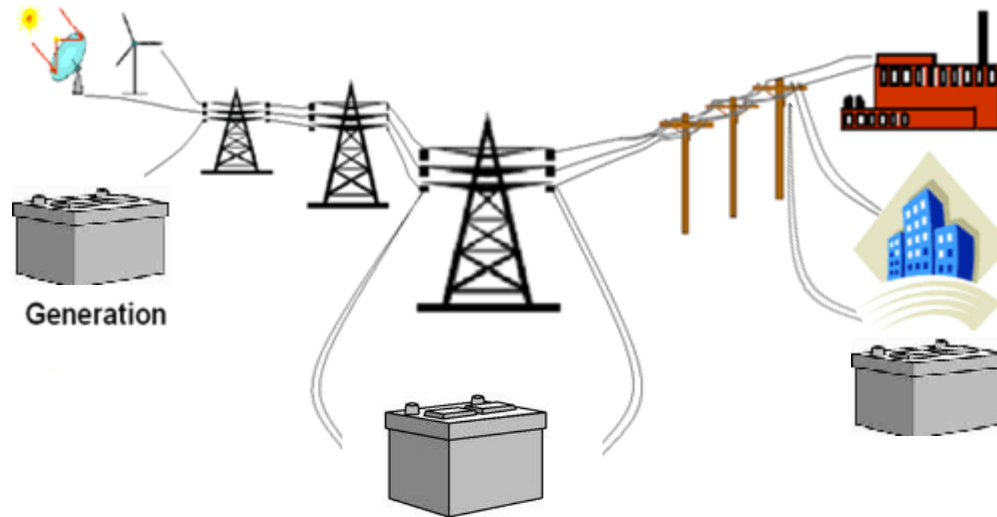
» Store Energy Electro-magnetically

- Superconducting magnetic energy storage (SMES)
- Super-capacitor

» Store Energy Electro-chemically

- Lead-acid battery
- Flow batteries
- Other advanced batteries

Energy Storage System

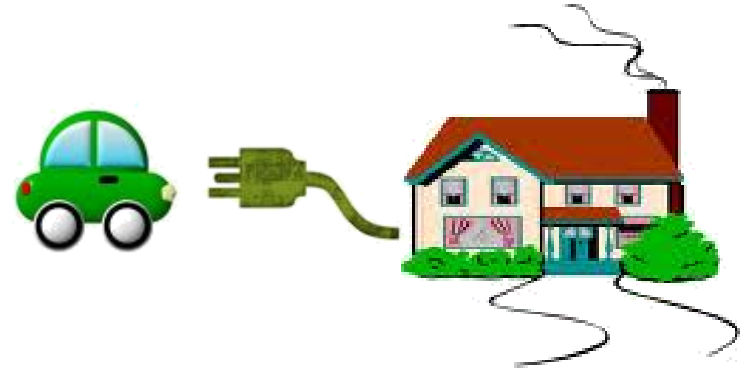


► Roles

- Supply energy for small-sized power grid
- Support renewable energy conversion system
- Provide backup power to prevent outages
- Improve power quality
- Match electricity supply to load demand

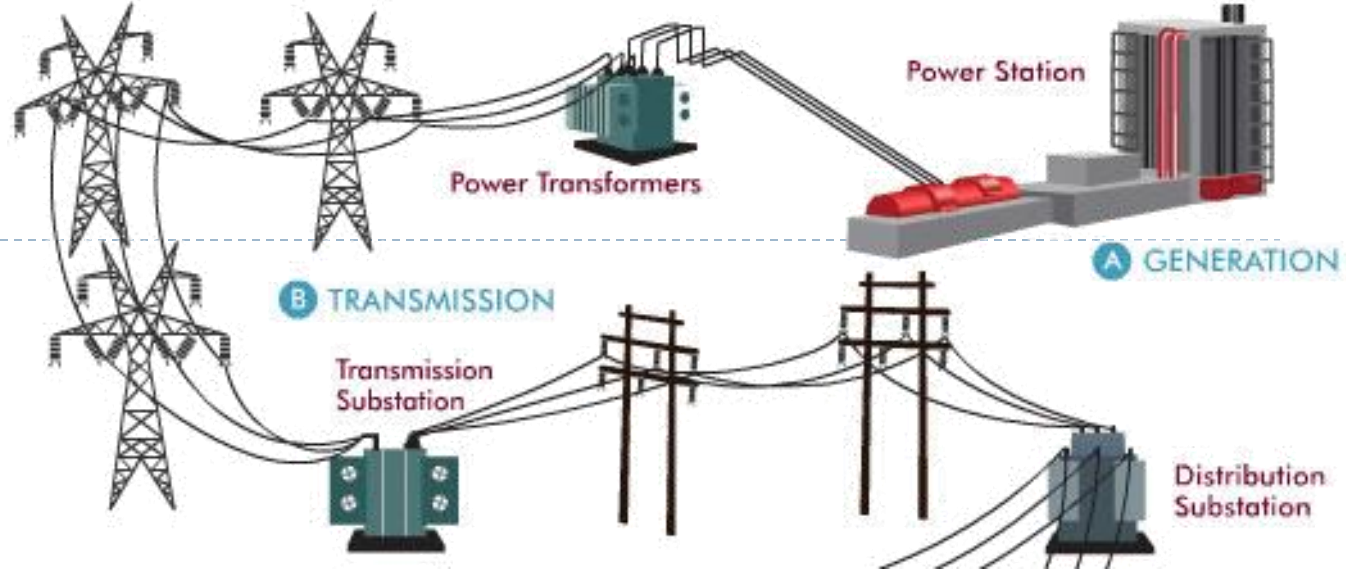
Electric Vehicle

- ▶ EV to Grid (V2G)
 - ▶ Leveling peak load
 - ▶ Backup for power failure

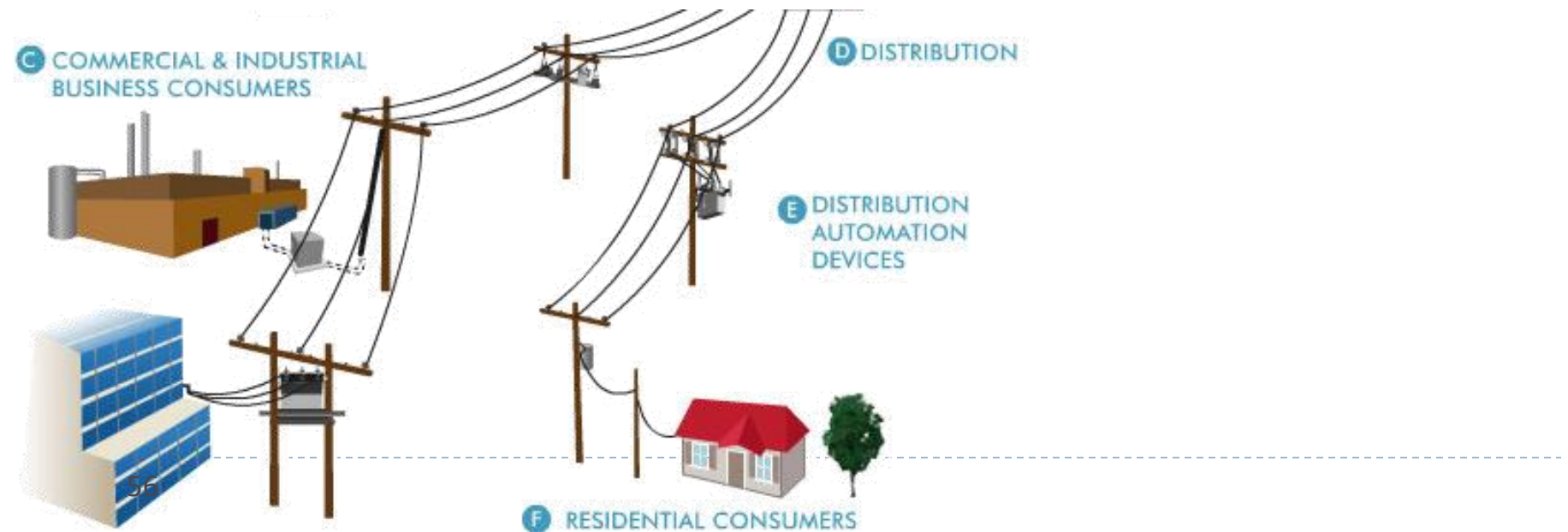


- ▶ Issues need to be solved
 - ▶ High penetration may affect the power system operation.
 - ▶ Charging spots and replacing station for drain batteries are needed.
 - ▶ Lack of standards and solutions for reverse power flow from load to T&D.



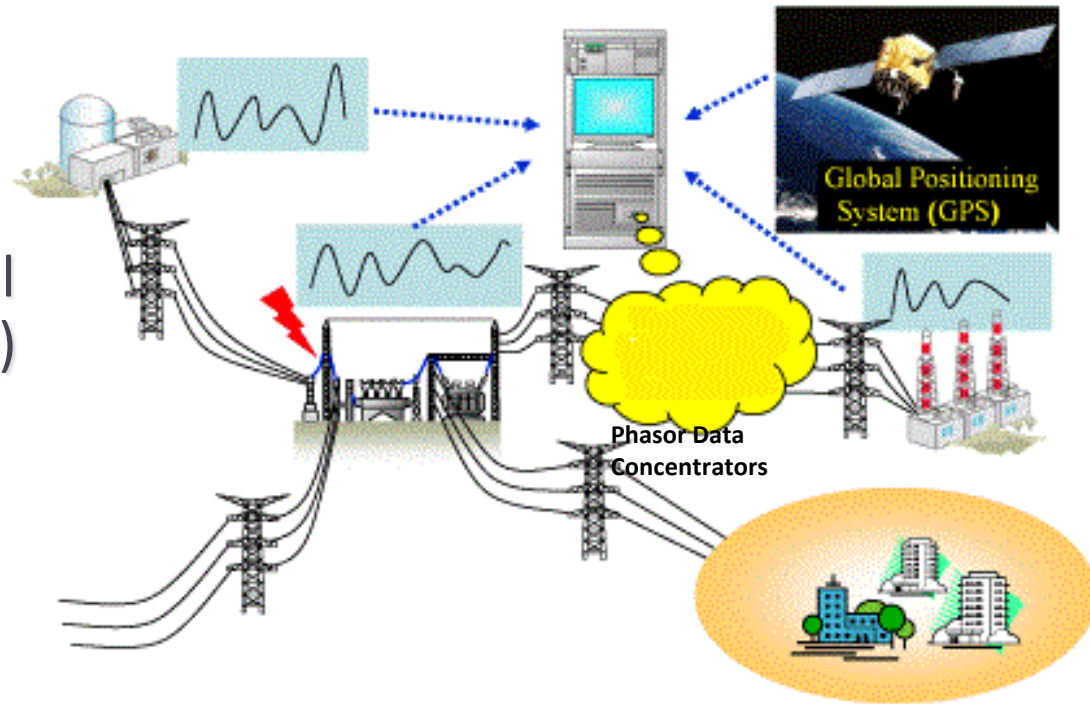


Smart Transmission and Distribution System

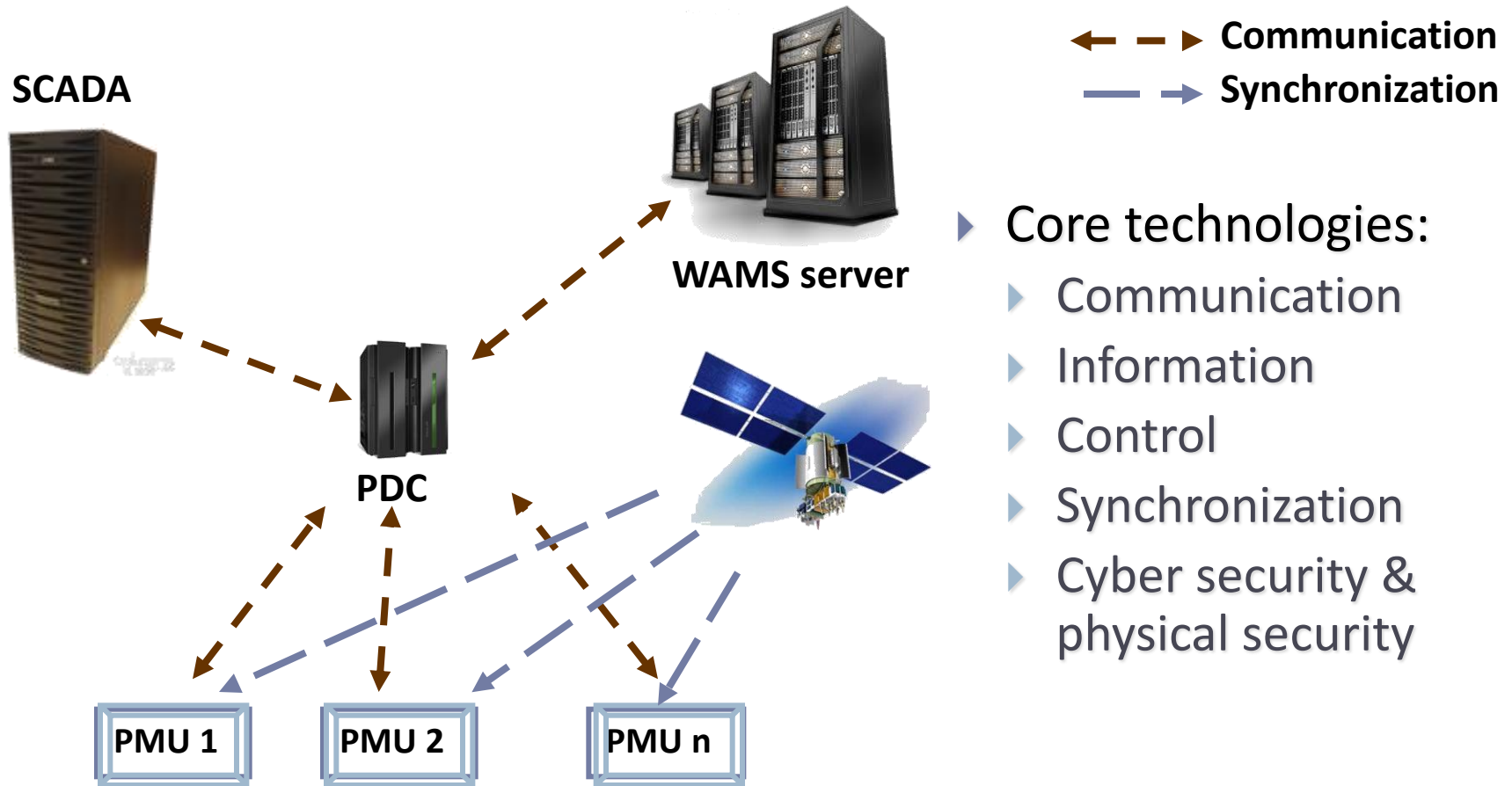


Phasor Measurement Unit (PMU)

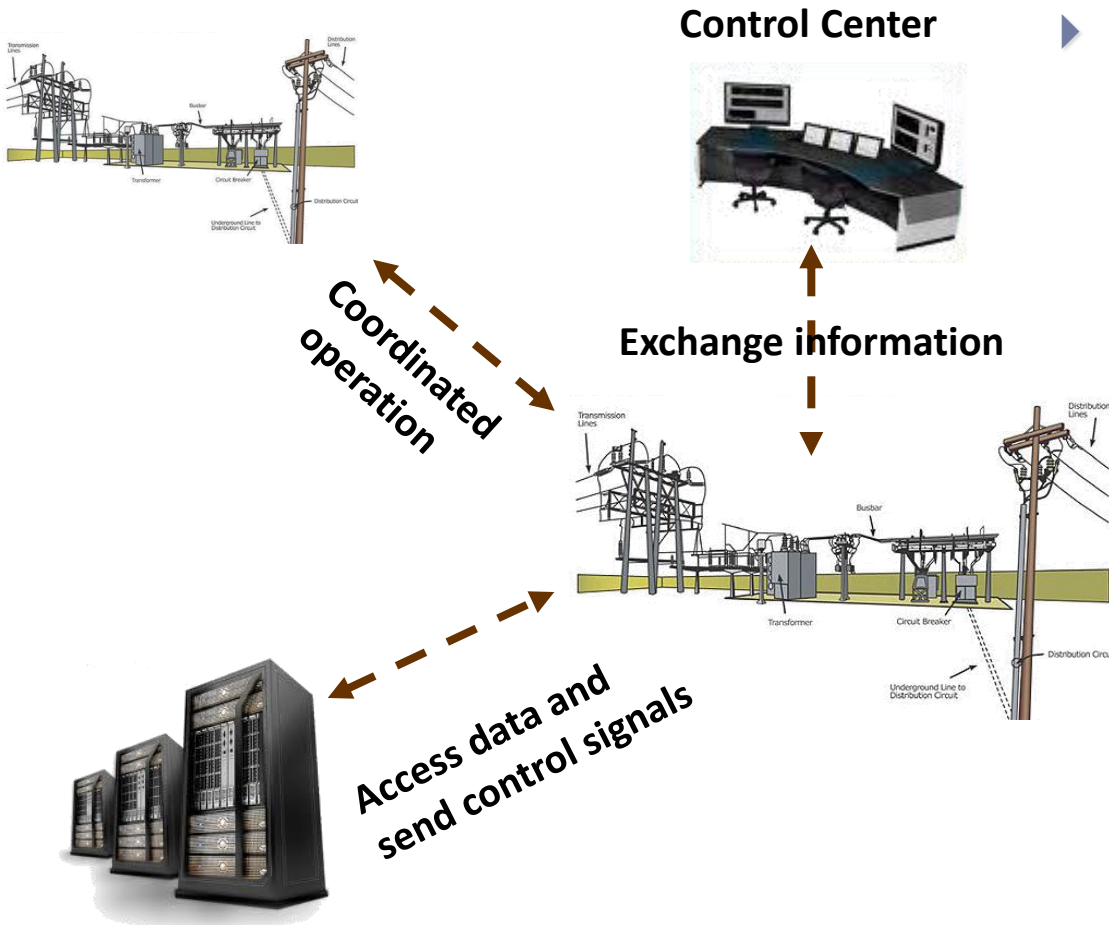
- ▶ A PMU measures electrical waves on an electricity grid.
- ▶ Installed in widely dispersed locations
- ▶ Synchronized by a global positioning system (GPS) radio clock
- ▶ A phasor data concentrators (PDC) collects data from PMUs and sends data to SCADA and WAMS server.



Wide Area Measurement System (WAMS)



Smart Substation Automation



► Core technologies

- Integrated asset management functionality
- Standard intelligent process interfaces (IEC 61850)
- Standard communication and processes inside the station and among stations
- Digital system, online information, and intelligent applications
- Digital protection devices and components for online network analyses

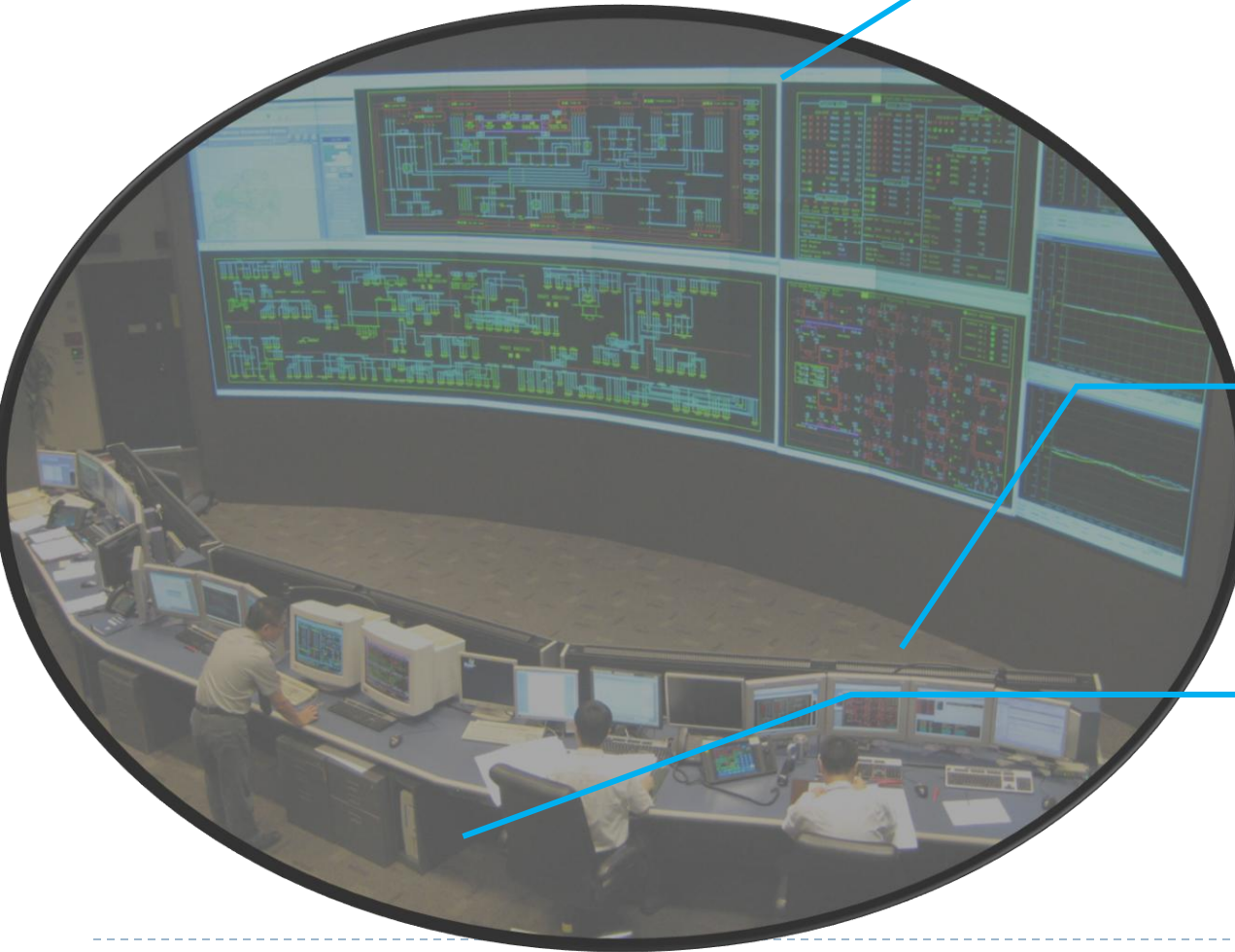
WAMS

Smart Control Center

- ▶ A Smart Control Centre is the advanced operational centre that coordinates, monitors, and optimizes the entire electricity network:
 - ▶ Real-time data acquisition and processing from sensors, smart meters, and devices across the network
 - ▶ Advanced visualization systems that transform complex grid data into actionable information for operators
 - ▶ Automated decision-making algorithms that can detect anomalies, predict issues, and implement corrective measures
 - ▶ Integration of artificial intelligence and machine learning for grid optimization, load forecasting, and predictive maintenance
 - ▶ Coordination of distributed energy resources, storage systems, and conventional generation to maintain grid balance
 - ▶ Dynamic management of power flows to prevent overload and optimize transmission efficiency
 - ▶ Cybersecurity monitoring and response capabilities to protect critical infrastructure
 - ▶ Demand response programme management to adjust consumer loads during peak periods



Control Center



Visualization techniques:

- Artificial intelligence-driven data reduction
- Color contouring and animation
- Rapid refresh
- Voice recognition
- Virtual reality environments

Decision support analysis:

- Advanced human computer interaction
- Risk-based decision-making algorithms

operational training & “what-if” analysis:

- Advanced programs
- Advanced simulators

Core Technologies

Integrated two-way communications

Control center

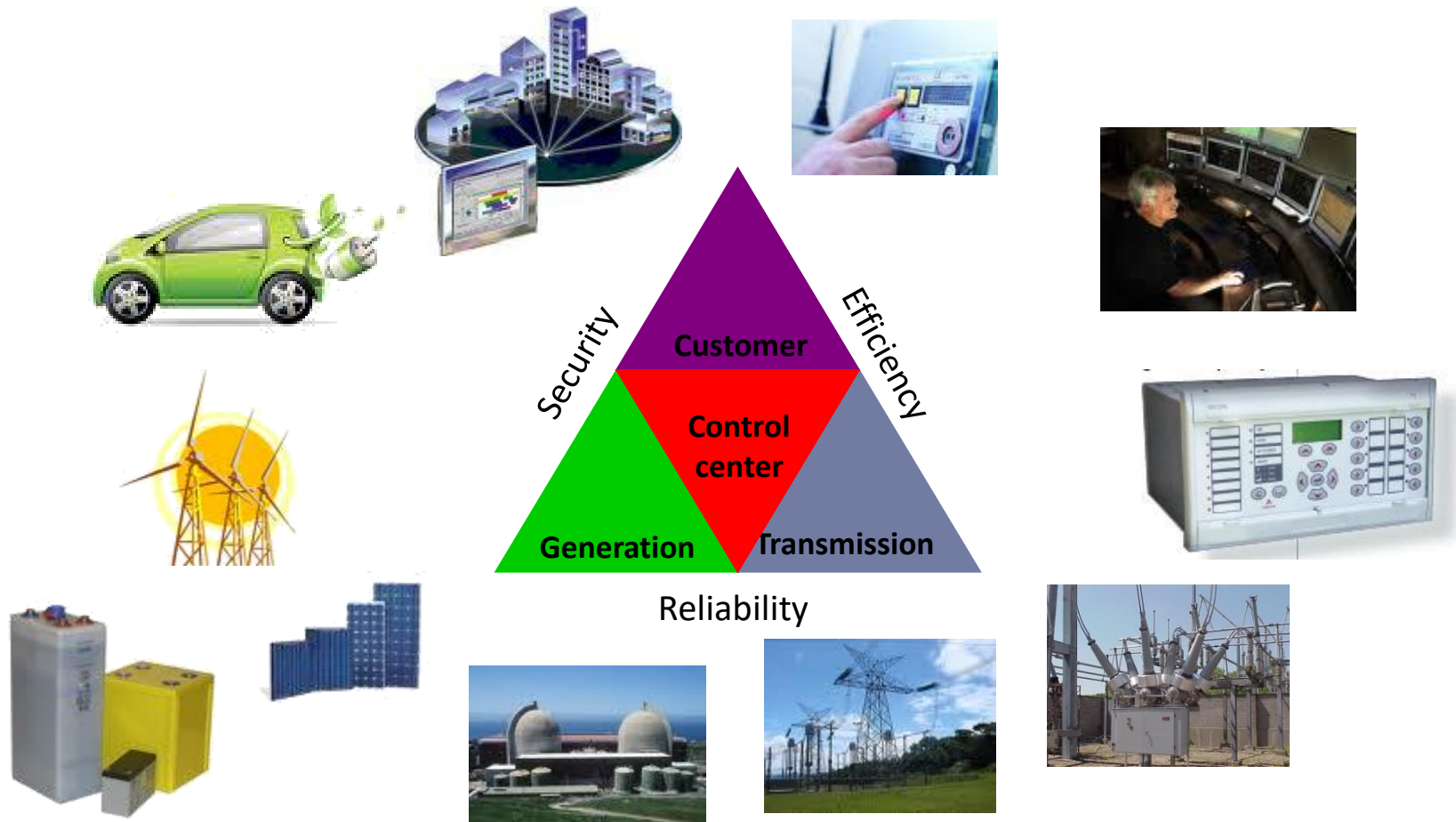
Improved Interfaces and Decision Support

- Visualization techniques
- Decision support analysis
- Operational training & “what-if” analysis

Advanced control methods

- ▶ Collect data and monitor grid components
- ▶ Analyze data
- ▶ Detect faults
- ▶ Take autonomous action when appropriate
- ▶ Provide information and options for human operators
- ▶ Integrate with other enterprise-wide processes and technologies

Sustainability of Smart Grid



Challenges for the operation of traditional grid

- ▶ Penetration of renewable energy resources (RERs); bidding strategies of participants prevent companies from providing solutions with environmental goals.
- ▶ High dependency of power system models on intelligent operation and control, power system planning and control prevent companies from providing solutions with technical goals.
- ▶ Complexity in operation and control of power system due to the complexity in computational tools used for modeling and uncertainties.
- ▶ Forecasting of load demand, price and ancillary services prevent the companies from providing solutions with economic goals.
- ▶ Increased distributed generations (DGs) and demand response (DR) in electric market; and tuning of controller parameters in varying operating conditions prevent the electric utility companies from providing smart solutions with economic, environmental, and technical goals.
- ▶ Risk minimization in electric power sector with investment in computational tools seeks to determine trade-off between maximizing the expected returns.